

PEDIATRIC CHEST X-RAY CLASSIFICATION VIA CONTRASTIVE DOMAIN ADAPTATION FROM ADULT DATA

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ABSTRACT

Radiology AI models are trained almost exclusively on adult chest X-rays (CXRs), leaving pediatric diagnosis severely underserved. Pediatric CXRs differ from adult CXRs in anatomy, disease presentation, and imaging protocols, yet labeled pediatric data is scarce. We frame this as a domain adaptation problem: leverage abundant labeled adult CXRs to improve pneumonia classification on a small labeled pediatric dataset. Our approach first performs contrastive pre-training on 112,120 adult CXRs from NIH ChestX-ray14, aligning image features with class-level text prompts via an InfoNCE loss inspired by CLIP . We then fine-tune the image encoder on 5,863 pediatric CXRs using a contrastive alignment loss anchored to the frozen adult-learned text embeddings. Classification is performed by nearest-neighbor lookup in the shared text-image latent space. We evaluate on AUC-ROC and F1, vary the pediatric fine-tuning set size to produce performance curves, and visualize latent space geometry via t-SNE before and after adaptation. We compare against a pediatric-only baseline and an alternative domain alignment strategy. Our results address a central question: does adult contrastive pretraining help, and under what conditions does the performance gap with the baseline close? Our code is available at: <https://github.com/eb45/Pediatric-CXR-Adapt>

1 INTRODUCTION

Pneumonia remains one of the leading causes of mortality among children worldwide, accounting for a disproportionate burden of preventable deaths in the pediatric population. Thus, early and accurate diagnosis is critical to improving outcomes. Chest X-ray (CXR) imaging has emerged as the frontline diagnostic tool for pneumonia detection: it is widely available, low-cost, rapid to acquire, and provides sufficient anatomical detail to distinguish pneumonia from normal lung tissue. However, interpreting pediatric CXRs is non-trivial, as the developing thorax presents anatomical features that differ substantially from adult radiographs, and access to trained radiologists is limited in many low-resource settings where pediatric pneumonia burden is highest.

A natural approach to this problem is to leverage deep learning models trained on large, well-annotated CXR datasets. Adult CXR datasets contain hundreds of thousands of labeled images and have enabled strong diagnostic models. Pediatric datasets, by contrast, are far smaller and more sparsely annotated, making it difficult to train high-performing models from scratch on pediatric data alone. This discrepancy motivates a *domain adaptation approach*: train a model on the rich adult domain and transfer it to the pediatric domain.

The key insight motivating our approach is that one should learn a representation that captures pathology semantics (pneumonia vs. normal) in the adult domain and then align pediatric images to that representation, rather than re-learning from scratch on limited pediatric data.

Contributions. We (1) adapt contrastive vision-language pretraining to the adult-to-pediatric CXR domain shift, (2) propose and justify an alternative domain alignment strategy, (3) conduct a systematic ablation over pediatric fine-tuning set size, and (4) provide t-SNE visualizations that directly test whether domain alignment occurs.

2 RELATED WORK

CXR datasets

PEDIATRIC DATASET (TARGET DOMAIN) The Kermany et al. (2018) Pediatric Chest X-ray Dataset consists of 5,863 labeled chest X-ray images of children aged 1-5 years, collected from pediatric patients and annotated for pneumonia vs. normal classification. The dataset is relatively small compared to adult datasets but is clinically important as pediatric chest X-rays exhibit distinct anatomical characteristics (e.g., smaller thoracic structures, developing lungs) and different disease manifestations. These differences, combined with limited data availability, make pediatric CXR classification a challenging task, motivating its use as the target domain in this work.

ADULT DATASET (SOURCE DOMAIN) The NIH Chest X-ray14 (Wang et al. (2017)) dataset contains 112,120 frontal-view chest X-ray images from over 30,000 adult patients, with labels for 14 thoracic diseases. In this project, the dataset is simplified to a binary classification task (pneumonia vs. normal) to align with the pediatric dataset. Due to its large scale and diversity, this dataset provides a rich source of labeled data for representation learning, making it well-suited as the source domain for pretraining. However, models trained solely on this dataset often fail to generalize to pediatric data due to domain shift in anatomy, pathology, and imaging conditions.

Domain adaptation

Domain adaptation addresses the problem of distribution shift between training (source) and deployment (target) data, where the underlying data distributions differ despite sharing the same task. Formally, a domain is defined by a data distribution and labeling function. The goal of domain adaptation is to learn a model on the source domain that generalizes well to the target domain by minimizing the target error despite this mismatch.

A foundational approach is the Domain-Adversarial Neural Networks (Ganin et al. (2016)) framework, which introduces adversarial training to learn domain-invariant feature representations. The model jointly optimizes a task-specific classifier while using a domain discriminator to encourage features that cannot distinguish between source and target domains, effectively reducing domain

shift. This aligns with the general domain adaptation objective of finding a transformation that brings source and target distributions closer, enabling knowledge transfer when their probability distributions differ.

Contrastive and self-supervised learning

Contrastive learning is a self-supervised learning paradigm that learns representations by bringing similar (positive) pairs closer and pushing dissimilar (negative) pairs apart in a shared embedding space. This approach enables models to learn meaningful features from unlabeled data, which is particularly valuable in medical imaging where annotations are scarce.

A foundational framework is SimCLR (Chen et al. (2020)), which demonstrates that strong visual representations can be learned by contrasting augmented views of the same image without requiring labels. SimCLR highlights the importance of data augmentations and large batch sizes, establishing contrastive learning as a powerful alternative to supervised pretraining.

Building on this paradigm, CLIP extends contrastive learning to multimodal settings, aligning images and text in a shared embedding space using large-scale image-caption pairs. This enables zero-shot classification, where models can generalize to new tasks using natural language prompts without additional training. Radford et al. (2021)

In the medical domain, CheXzero (Tiu et al. (2022)) adapts CLIP-style training to chest X-rays by leveraging paired radiology reports, enabling zero-shot pathology classification without explicit labels. This approach achieves performance comparable to expert radiologists and demonstrates that contrastive vision-language models can learn clinically meaningful representations directly from medical data.

3 METHODOLOGY

3.1 DATA

We use two publicly available datasets summarized in Table 1.

The adult source domain is NIH ChestX-ray14, collection of 112,120 frontal-view CXRs with disease labels extracted from radiology reports via NLP. We simplify the original 14-class annotation to a binary task: images with `Pneumonia` in their finding labels are assigned $y = 1$; all remaining images are assigned $y = 0$. We use the dataset’s official train/test split provided by `train_val_list.txt` and `test_list.txt`.

The pediatric target domain is the Kermany et al. dataset, comprising of CXRs from children aged 1–5 with binary labels (Pneumonia / Normal), available on Kaggle. The dataset is organized into `train/` and `test/` splits with class subfolders (`NORMAL/`, `PNEUMONIA/`); as no validation folder is present, we carve out a stratified 10% of the training set (stratified by class) as a validation split for hyperparameter tuning.

All images are resized to 224×224 pixels via bicubic interpolation and converted to RGB. Pixel values are normalized using ImageNet statistics (mean = (0.485, 0.456, 0.406), std = (0.229, 0.224, 0.225)). Text prompts for the two classes are tokenized using `bert-base-uncased` with a maximum sequence length of 64 tokens. A fixed random seed of 42 is used throughout all data preparation steps to ensure reproducibility.

Dataset	Domain	Total	Pneumonia	Normal
NIH ChestX-ray14	Adult (source)	112,120	~1,431	~60,361
Kermany et al.	Pediatric (target)	5,863	4,273	1,590

Table 1: Dataset statistics for source (adult) and target (pediatric) domains.

3.2 METHOD

We train a dual-encoder model on NIH ChestX-ray14. The image encoder $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ maps a CXR image to a d -dimensional embedding. The text encoder $g_\phi : \mathcal{T} \rightarrow \mathbb{R}^d$ maps a natural-language prompt to the same embedding space. We compare using ResNet-18 and ResNet-50 as the image encoders and a pretrained transformer as the text encoder, both initialized from open-source CLIP weights.

For each binary label $y \in \{\text{PNEUMONIA}, \text{NORMAL}\}$, we define a class-level text prompt:

- “A chest X-ray showing pneumonia.”
- “A normal chest X-ray with no findings.”

We optimize the CLIP-style loss over mini-batches of (image, label) pairs, treating the corresponding text class as the positive and all other classes in the batch as negative:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(I, T_y))}{\sum_{t \in \mathcal{T}} \exp(\text{sim}(I, t))}$$

Where I represents the image and T corresponds to the correct prompt. Both encoders are updated during this stage.

After pretraining, we freeze the text encoder g_ϕ and fine-tune only the image encoder f_θ on the labeled pediatric training set. The frozen text encoder provides stable class-level anchors in latent space; the image encoder is trained to map pediatric CXRs close to the appropriate anchor.

The fine-tuning loss is the same CLIP-style objective applied to pediatric (image, label) pairs and the frozen text embeddings. This forces the image encoder to map pediatric pneumonia images near $g_\phi(t_{\text{PNEUMONIA}})$ and pediatric normal images near $g_\phi(t_{\text{NORMAL}})$, even though the text encoder has never seen pediatric data. Crucially, the text embeddings act as a semantic bridge: because they were learned alongside adult pathology images, they encode pathology semantics that are label-consistent across domains.

At inference time we embed each image with f_θ and compare it to two *frozen* class text embeddings $g_\phi(t_k)$ (L2-normalized). Logits are

$$\ell_k = \tau \langle f_\theta(\mathbf{x}), g_\phi(t_k) \rangle = \tau \cos(f_\theta(\mathbf{x}), g_\phi(t_k)), \quad (1)$$

with learned temperature $\tau = \exp(\text{logit_scale}) > 0$ (CLIP-style). There is no extra classifier head; $\hat{y} = \arg \max_k \ell_k = \arg \max_k \cos(\cdot, \cdot)$, and $\text{softmax}(\ell)$ yields class probabilities.

3.3 EVALUATION METRICS

We report AUC-ROC and F1 score on the held-out pediatric test set ($N = 624$). To assess data efficiency, we vary the fraction of the pediatric training set available during fine-tuning and report the resulting performance curve. We visualize the latent space geometry via t-SNE before and after fine-tuning to verify whether domain alignment occurs. All metrics are computed over the pediatric test set; AUC-ROC confidence intervals are obtained by bootstrap resampling (2,000 samples).

4 RESULTS

4.1 T-SNE LATENT SPACE GEOMETRY EVALUATION

Before fine-tuning, Normal and Pneumonia embeddings are largely intermixed. After fine-tuning with text-anchored contrastive loss, the two classes form well-separated clusters, confirming that domain alignment occurs.

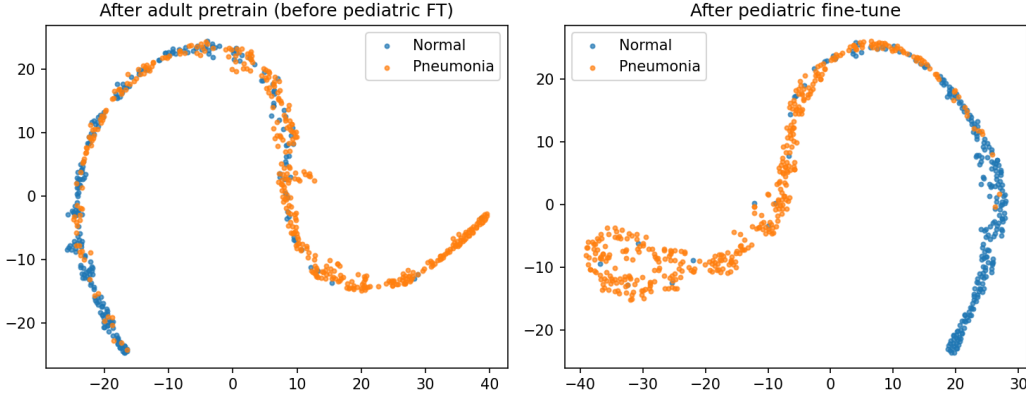


Figure 1: t-SNE of pediatric CXR embeddings before fine-tuning (left) and after pediatric fine-tuning (right) for the ResNet-18 encoder

4.2 RESNET-18 VERSUS RESNET-50

We evaluate two image encoder backbones—ResNet-18 and ResNet-50 to assess how encoder capacity affects domain adaptation performance on the pediatric test set. Below in Table 2, we see that ResNet-50 achieves substantially higher AUC-ROC (0.9394 vs. 0.8375) and near-perfect sensitivity (0.997), making it the preferred encoder for clinical deployment where missing a pneumonia case is costly. ResNet-18 achieves better specificity (0.496 vs. 0.286) and a more balanced operating point at the default threshold of 0.5, which may be preferable in settings where false positives carry high cost. The ResNet-50 model outperforms the baseline but the ResNet-18 doesn’t highlighting that adult pre-training is not unconditionally beneficial: the image encoder must have sufficient representational capacity to transfer the learned pathology semantics.

Table 2: Pediatric test set performance ($N = 624$). CI = 95% bootstrap (2,000 samples).

Model	AUC-ROC	AUC CI	F1 (0.5)	Sensitivity	Specificity	Best F1
Baseline (pediatric-only, ResNet-50)	0.8745	[0.8169, 0.9320]	0.8090	X	X	0.8588
Ours, ResNet-18	0.8375	[0.803, 0.870]	0.8349	0.933	0.496	0.844
Ours, ResNet-50	0.9394	[0.916, 0.960]	0.8224	0.997	0.286	0.912

In Figure 2 the proposed model—an adult-pretrained image–text model with pediatric fine-tuning using ResNet-50—is compared to a pediatric-only baseline with no adult pretraining, evaluated for training using 1%, 25%, 50%, and 100% of the pediatric training data. The proposed model consistently achieves higher AUC-ROC and F1 scores across all settings. The performance gap is largest at lower data fractions, indicating that adult pretraining provides a strong initialization and improves sample efficiency. While both models improve as more pediatric data is available, the proposed method remains more stable and maintains its advantage, suggesting better generalization in the low-data regime.

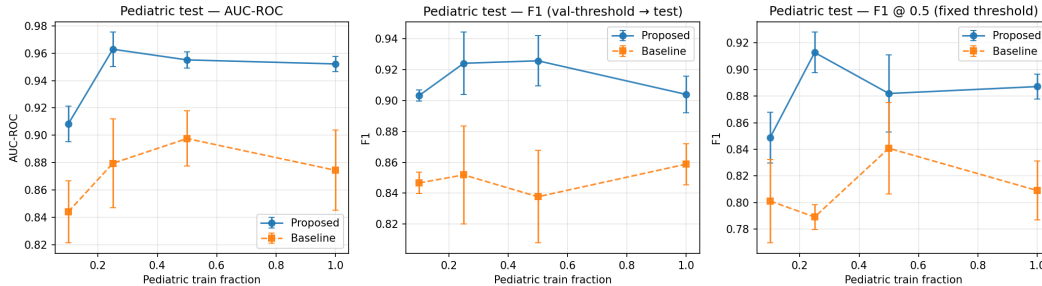
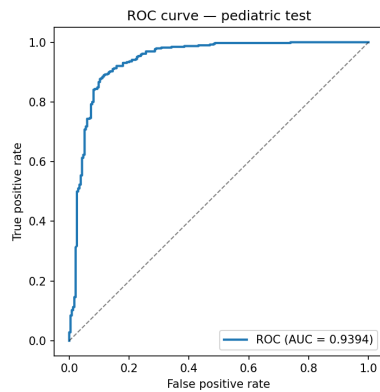
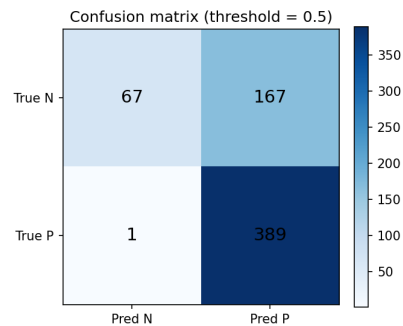


Figure 2: Performance curves for ResNet-50

ResNet-50 results. Figure 3a shows the ROC curve. The steep initial rise indicates high sensitivity is achieved at low false-positive rates, making this model well-suited to clinical screening where missed diagnoses are the primary concern. Figure 3b shows the raw counts for the confusion matrix for ResNet-50 at threshold 0.5. Of 390 true pneumonia cases, only 1 is missed and of 234 true normal cases, 167 are classified as pneumonia.



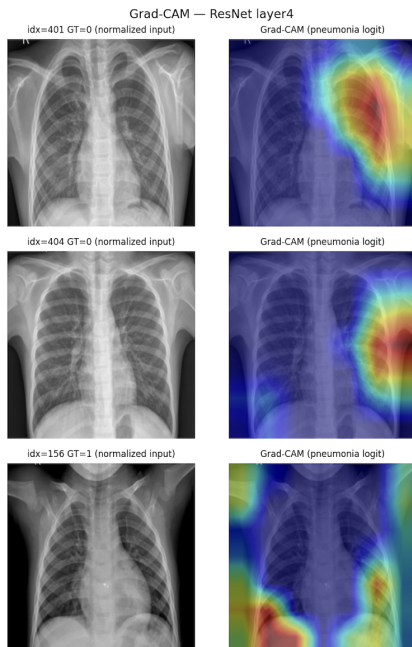
(a) ROC curve for ResNet-50



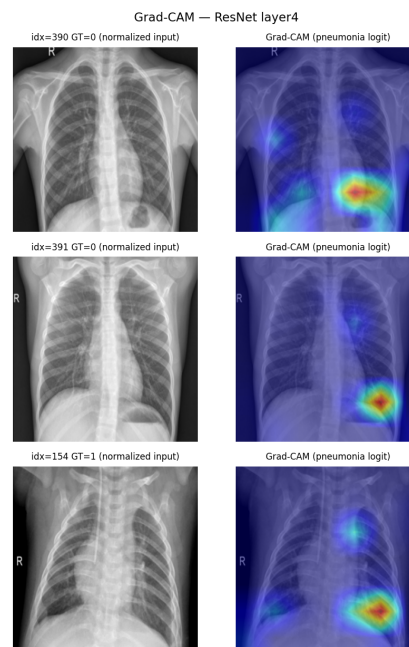
(b) Confusion matrix

Figure 3: ResNet-50 performance

Grad-CAM Saliency Map Visual Differences. We compare class-discriminative saliency for pediatric CXR using Grad-CAM on layer 4 (last conv block before pooling) with respect to the pneumonia logit, for ResNet-18 (Figure 4a) and ResNet-50 (Figure 4b). On the same-style normalized inputs, ResNet-18 tends to produce broader, more diffuse heatmaps over large lung fields. ResNet-50 yields tighter, more localized highlights, often concentrated in basal or focal lung regions.



(a) Grad-CAM for ResNet-18



(b) Grad-CAM for Resnet-50

Figure 4: Grad-CAM Saliency Map Comparison

4.3 ALTERNATIVE STRATEGY: ALIGNING ADULT AND PEDIATRIC REPRESENTATIONS IN LATENT SPACE

While the contrastive text-anchor approach aligns domains implicitly through shared semantic embeddings, the alternative strategies explored (DANN and Optimal Transport) perform domain alignment explicitly in feature space.

DANN enforces domain invariance via adversarial training, encouraging features that cannot distinguish between adult and pediatric domains. In addition to the main task (e.g., pneumonia classification), a domain classifier is introduced to predict whether a feature originates from the source or target domain. A gradient reversal layer is used so that the feature extractor is trained to maximize the domain classifier’s error while still minimizing classification loss. This results in learned feature representations that are discriminative for the task but indistinguishable across domains, effectively reducing domain shift.

Optimal Transport (Wasserstein) directly minimizes the distributional distance between source and target embedding distributions. Given source and target feature distributions, OT seeks a transport plan that maps one distribution onto the other with minimal cost. This cost is typically defined in terms of a ground metric. The resulting metric, the Wasserstein distance (Earth Mover’s Distance), can be interpreted as the minimum “effort” required to transform one distribution into another. Courty et al. (2016)

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} d(x, y)^p d\gamma(x, y) \right)^{\frac{1}{p}} \quad (2)$$

These methods serve as meaningful alternatives because they do not rely on text supervision or semantic grounding, but instead attempt to resolve domain shift purely through distribution alignment mechanisms. This allows us to evaluate whether explicit feature-level alignment can match or outperform semantic contrastive alignment, particularly in the presence of limited labeled pediatric data.

4.3.1 ALT STRATEGY 1: DANN

As an alternative domain alignment strategy, we implement a Domain-Adversarial Neural Network (DANN). Rather than relying solely on text-anchored contrastive fine-tuning to bridge the adult-to-pediatric gap, DANN explicitly encourages the image encoder to produce *domain-invariant* representations by training a domain discriminator adversarially against the feature extractor. A gradient reversal layer (GRL) is inserted between the encoder and discriminator: during backpropagation, gradients from the domain classifier are negated, pushing the encoder toward representations that are indistinguishable between adult and pediatric domains while remaining discriminative for the pneumonia classification task.

Results. Table 3 reports DANN performance on the pediatric test set ($N = 624$). DANN achieves AUC-ROC of 0.9242 (95% CI [0.897, 0.947]), F1 of 0.860 at threshold 0.5, and a best F1 of 0.905 at threshold 0.9875. Sensitivity is near-perfect (0.997) at the default threshold, with specificity of 0.462. Figure 5 shows the ROC curve.

Model	AUC-ROC	AUC CI	F1 (0.5)	Sensitivity	Specificity	Best F1
ResNet-50 (text-anchor)	0.9394	[0.916, 0.960]	0.822	0.997	0.286	0.912
DANN (ResNet)	0.9242	[0.897, 0.947]	0.860	0.997	0.462	0.905

Table 3: DANN pediatric test set performance

Comparison to text-anchor approach. Both methods achieve near-identical sensitivity (0.997); however, DANN produces a more balanced operating point at the default threshold, with notably higher specificity (0.462 vs. 0.286). The text-anchor approach retains a higher AUC-ROC (0.9394 vs. 0.9242), suggesting that semantic grounding via frozen text embeddings provides a stronger alignment signal than adversarial domain confusion alone across all classification thresholds.

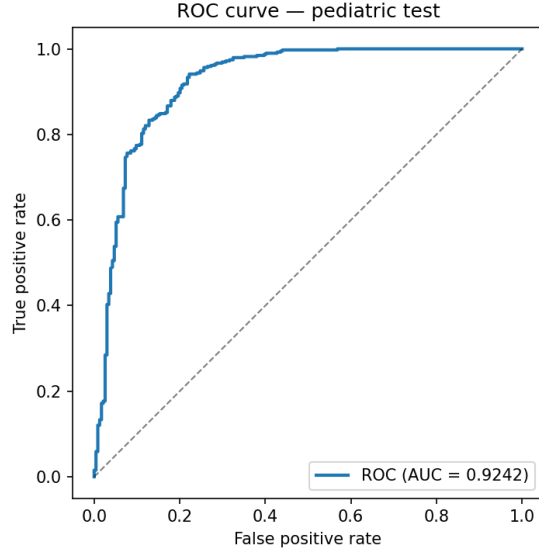


Figure 5: ROC curve for DANN on the pediatric test set

4.3.2 ALT STRATEGY 2: OPTIMAL TRANSPORT (WASSERSTEIN)

As a second alternative domain alignment strategy, we implement Optimal Transport (OT) using the Wasserstein distance to explicitly align feature distributions between adult and pediatric domains. In contrast to our primary contrastive, text-anchored alignment approach—which leverages semantic supervision from text embeddings—OT operates purely in feature space and does not rely on text supervision. Unlike DANN, which enforces domain invariance through adversarial training, OT provides a geometrically grounded alignment objective, encouraging the encoder to preserve meaningful structure while reducing domain shift. This makes OT a complementary alternative to both adversarial alignment (DANN) and semantic alignment (contrastive text-anchor).

In our implementation, embeddings from the adult (source) and pediatric (target) domains are aligned by minimizing the Wasserstein distance in feature space. This allows the model to leverage the global geometry of both distributions.

Results. Table 4 reports OT performance on the pediatric test set ($N=624$). The OT model achieves an AUC-ROC of 0.9508 (95% CI [0.9325, 0.9672]), outperforming both the baseline text-anchor and DANN approaches. At the default threshold (0.5), the model attains an F1 score of 0.8571, with perfect sensitivity (1.0000) and improved specificity (0.4444) compared to the text-anchor baseline. The best F1 score is 0.8912, indicating strong overall classification performance. Figure 6 shows the ROC curve.

Model	AUC-ROC	AUC CI	F1 (0.5)	Sensitivity	Specificity	Best F1
ResNet-50 (text-anchor)	0.9394	[0.916, 0.960]	0.822	0.997	0.286	0.912
OT (Wasserstein)	0.9508	[0.9325, 0.9672]	0.8571	1.0000	0.4444	0.8912

Table 4: OT pediatric test set performance

Comparison to other approaches. Compared to both DANN and the text-anchor method, OT achieves the highest AUC-ROC, suggesting that explicit distribution matching provides a stronger alignment between domains. While all methods maintain extremely high sensitivity, OT improves specificity over the text-anchor baseline and remains competitive with DANN. These results indicate that distribution-level alignment via Wasserstein distance can outperform adversarial domain confusion, while also avoiding instability associated with adversarial training.

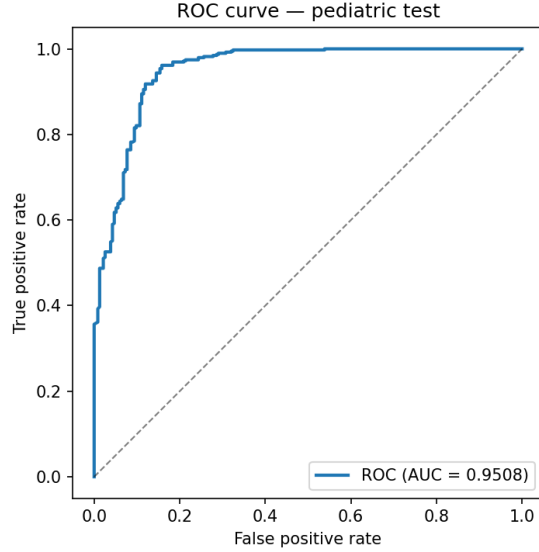


Figure 6: ROC curve for OT on the pediatric test set

5 CONCLUSION

In this work, we frame pediatric X-ray (CXR) classification as a domain adaptation problem, leveraging large-scale labeled adult data to address the scarcity of pediatric annotations. Our results demonstrate that adult contrastive pretraining significantly improves performance over a pediatric-only baseline, particularly in low-data regimes. The contrastive, text-anchored approach enables the model to learn semantically meaningful representations in the adult domain and transfer them efficiently to pediatric data, as evidenced by both improved AUC-ROC/F1 and clear class separation in t-SNE visualizations after fine-tuning.

We find that the benefit of adult pretraining is most pronounced when pediatric data is limited, where the baseline model struggles to learn robust representations. As the size of the pediatric fine-tuning set increases, the performance gap between the proposed method and the baseline narrows, suggesting that sufficient labeled pediatric data can partially compensate for the lack of pretraining. However, in realistic settings where pediatric data remains scarce, domain adaptation provides a critical advantage.

Across alignment strategies, the contrastive text-anchor approach achieves strong overall performance, indicating that semantic grounding via shared text embeddings is an effective mechanism for cross-domain transfer. At the same time, alternative strategies reveal important tradeoffs: DANN improves specificity through more balanced decision boundaries, while Optimal Transport achieves the highest AUC-ROC, demonstrating that explicit distribution alignment can further enhance performance. These results suggest that different alignment can further enhance performance. These results suggest that different alignment mechanisms capture complementary aspects of the domain shift.

Overall, our findings highlight that adult-to-pediatric transfer is not only feasible but highly effective when guided by appropriate alignment strategies. More broadly, this work underscores the importance of combining large-scale pretraining with principled domain adaptation techniques to improve generalization in data-scarce medical imaging settings.

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A APPENDIX

A.1 HYPERPARAMETER SETTINGS

Setting	Value
Image encoder	ResNet-18, ResNet-50
Text encoder	bert-base-uncased, max 64 tokens
Input resolution	224×224 , bicubic resize
Normalization	ImageNet mean/std
Embedding dimension	512
Batch size	16
Optimizer	AdamW, $\text{lr} = 10^{-4}$, weight decay = 10^{-4}
Epochs	5 (adult pretrain) + 5 (pediatric fine-tune)
Random seed	42

Table 5: Hyperparameter settings used during training.

A.2 TEXT PROMPT TEMPLATES

We use the following fixed text prompts for the two classes:

- PNEUMONIA: “A chest X-ray showing pneumonia.”
- NORMAL: “A normal chest X-ray with no findings.”